

1. Challenge

I choose to solve challenge 5 (Anomaly Detection for Thermal Drone Footage), applied to identify missing persons. I have tackled the approach beyond anomaly detection, as classifying which image is abnormal is far from sufficient. Ultimately, a rescue team needs to decide what path to take to find a missing person. Therefore, the output of my model is a path that maximizes the chance to find a missing person.

2. Tools / ML models you have used

2.1 Dataset

I used the provided dataset provided by the Black Forest Mountain Rescue Team. The dataset consists of 166 training images (all are normal without a missing person) and 60 test images (30 are normal and 30 are abnormal (with a missing person)). The images had no GPS location in their metadata, so I generated random GPS coordinates within the Black Forest in Germany for each image.

2.2 Modeling Approach

I tackled this challenge using a "predict-then-optimize" approach:

1. **Feature Extraction:** I extracted image embeddings using the penultimate layer of the pre-trained ResNet18 model.
2. **Prediction (anomaly detection):** After extracting embeddings from all 226 images, I used the k-nearest neighbors (kNN) algorithm to compute the mean distance of each test image to its 5 nearest neighbors in the training set.
3. **Optimization:** The image distance scores serve as an *input* to the optimization problem. The goal is to maximize the sum of image distance scores (proxy for the probability of finding a missing person), while constraining the total distance traveled by the rescue team (proxy for available time).

3. What has worked well with these tools?

The pre-trained ResNet18 model provided a powerful feature extractor, as it was trained on a large dataset (ImageNet) and is capable of capturing complex patterns in images. The optimization problem could also be solved efficiently using integer programming.

4. What was challenging?

The most challenging part is computing the distance scores for the test images. With 512 dimensions, the distance scores start suffering from the curse of dimensionality.

5. How have you spent your time?

Overall, I could only work for 5 hours on this challenge, as I feel quite sick and had to sleep for 10 hours overnight. I spent

- 30 minutes on understanding the challenge
- 3 hours on the technical implementation (1 hour for each of the three steps: feature extraction, prediction, and optimization)
- 1.5 hours on writing the report, summary and recording the videos